

ATP Selectively Releases ClpH, but Not GroX, from a Heat-Damaged Casein Matrix

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ABSTRACT

Extracts containing the fictional chaperones ClpH and GroX were applied to a matrix bearing heat-damaged casein. Both proteins bound the matrix. Addition of 1 mM ATP plus 5 mM MgCl₂ released 81% of bound ClpH but only 7% of bound GroX. GroX required pH 2.8 elution for recovery. ADP released 19% of ClpH, and ATP without MgCl₂ released 24%. No phosphorylation assay was performed. The results show nucleotide- and magnesium-dependent release of ClpH and a distinct retention behavior for GroX.

Keywords: ATP elution, casein affinity, ClpH, GroX, selective release

INTRODUCTION

Affinity matrices bearing damaged proteins can distinguish binding and release properties of chaperone-like proteins. Similar binding does not imply similar nucleotide-dependent dissociation.

This study compared ClpH and GroX in the same extract and on the same casein matrix. The main endpoint was the fraction of each bound protein released by ATP-Mg²⁺. Phosphorylation and transcriptional regulation were outside the study.

MATERIALS AND METHODS

Casein matrix

Casein was heated at 80 C for 20 min and coupled to agarose. Extracts were prepared in 25 mM Tris-HCl, pH 7.6, 150 mM NaCl, and 2 mM EDTA. Equal protein loads were applied to 1-mL columns.

Sequential elution

Columns were washed and then treated with 1 mM ATP plus 5 mM MgCl₂. Remaining proteins were eluted at pH 2.8. Separate columns received ADP plus MgCl₂ or ATP without MgCl₂.

Quantification

ClpH and GroX were quantified by immunoblotting. Percent release was calculated relative to the total amount recovered by nucleotide plus acid elution.

Elution condition	ATP/ADP	MgCl ₂
ATP-Mg	1 mM ATP	5 mM
ADP-Mg	1 mM ADP	5 mM
ATP alone	1 mM ATP	0
Acid	None	pH 2.8

Table 1. Sequential elution conditions.

RESULTS

Both proteins bound damaged casein

ClpH and GroX were both depleted from the flow-through relative to uncoupled agarose. Binding to native casein was less than 15% of binding to heat-damaged casein.

ATP-Mg2+ selectively released ClpH

ATP plus MgCl2 released 81 +/- 6% of bound ClpH but only 7 +/- 3% of GroX. Most GroX was recovered in the subsequent acid fraction.

Nucleotide and magnesium controls

ADP plus MgCl2 released 19 +/- 4% of ClpH. ATP without MgCl2 released 24 +/- 5%. Thus, efficient ClpH release required both ATP and MgCl2. No condition released more than 10% of GroX before acid elution.

Condition	ClpH released	GroX released
ATP + MgCl2	81 +/- 6%	7 +/- 3%
ADP + MgCl2	19 +/- 4%	6 +/- 2%
ATP, no MgCl2	24 +/- 5%	8 +/- 3%
Acid after ATP-Mg	remaining	major fraction

Table 2. Differential nucleotide-dependent release.

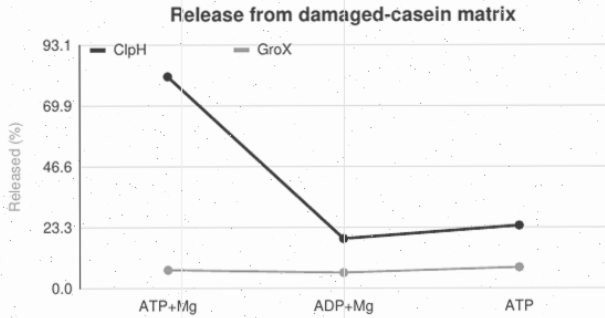


Figure 1. ATP-Mg2+ efficiently released ClpH but not GroX.

DISCUSSION

ClpH and GroX shared affinity for damaged casein but differed sharply in release behavior. ClpH dissociation was strongly dependent on ATP and magnesium, whereas GroX remained bound until acid elution.

The experiment does not determine whether ATP hydrolysis, nucleotide binding, or a conformational transition caused ClpH release. It also provides no evidence of phosphorylation. References to kinases in the bibliography are contextual distractors only.

LIMITATIONS

The matrix used damaged casein rather than native cellular clients. ATPase activity and nucleotide occupancy were not measured directly.

CONCLUSION

ATP-Mg2+ released 81% of matrix-bound ClpH but only 7% of GroX, demonstrating distinct release mechanisms for two proteins that both bind heat-damaged casein.

REFERENCES

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